

Burmese Grapheme to Phoneme NMT with Marian

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AI Engineering Class (Foundation)

Overview

2 architectures x 3 variants

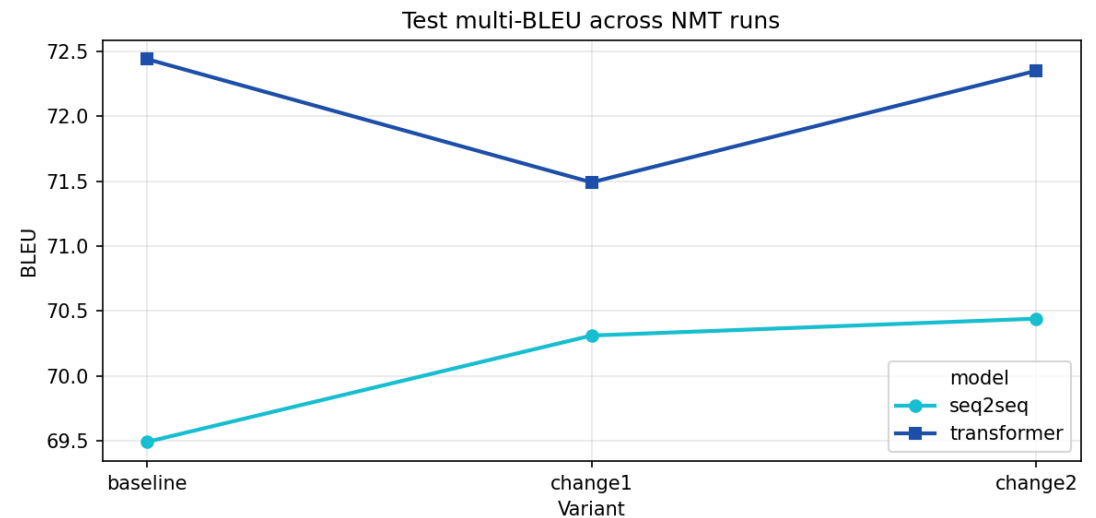
- 2 architectures
 - Sequence-to-sequence (Seq2Seq) = **seq2seq.mypsh.sh**
 - Transformer = **transformer.mypsh.sh**
- 3 variants
 - baseline → same configurations as Sayar's
 - change1 → more frequent checkpointing
 - change2 → checkpoint averaging around the peak via ensemble decoding (combines several model checkpoints at decode time by merging their output scores)
- Total = $2 \times 3 = 6$ end-to-end NMT runs

6 end-to-end NMT runs – 6 notebooks

- **nmt_seq2seq_v1.ipynb** = baseline seq2seq architecture
- **nmt_seq2seq_v2.ipynb** = change1 seq2seq architecture
- **nmt_seq2seq_v3.ipynb** = change2 seq2seq architecture
- **nmt_transformer_v1.ipynb** = baseline transformer architecture
- **nmt_transformer_v2.ipynb** = change1 transformer architecture
- **nmt_transformer_v3.ipynb** = change2 transformer architecture

Multi-BLEU on Test dataset across runs

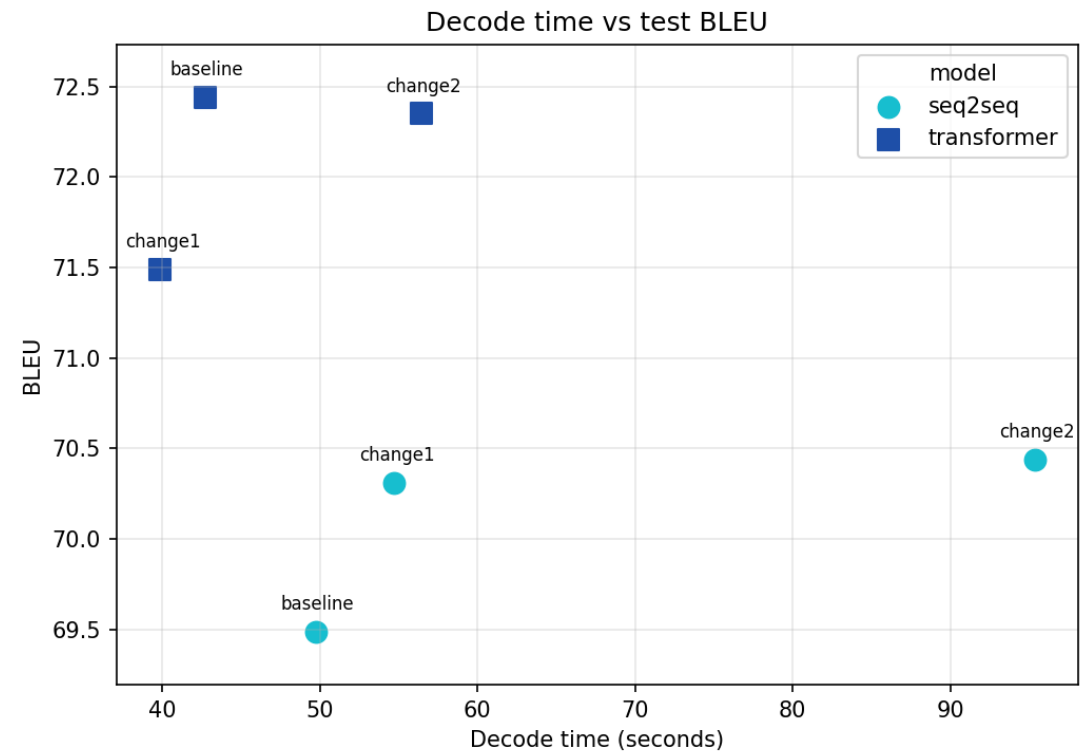
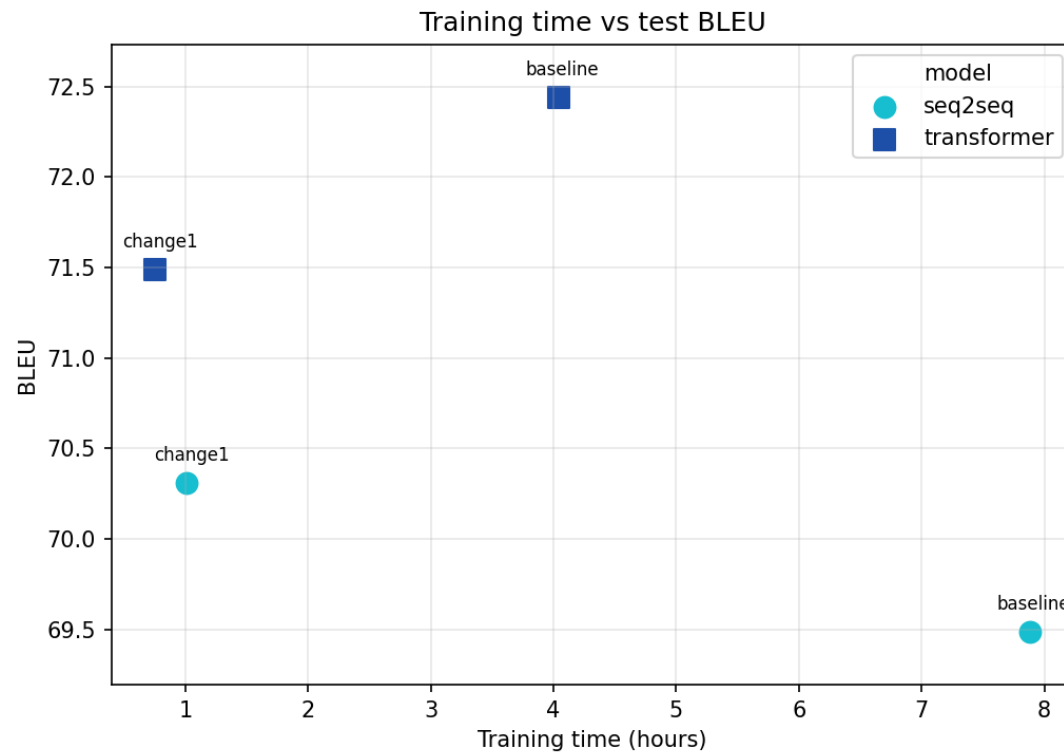
- baseline → same configurations as Sayar's
- change1 → set *--valid-freq* & *--save-freq* to 1000 from 5000; *--early-stopping* 5 from 10
- change2 → pass multiple models to *--models* with corresponding *--weights* to Marian decoder



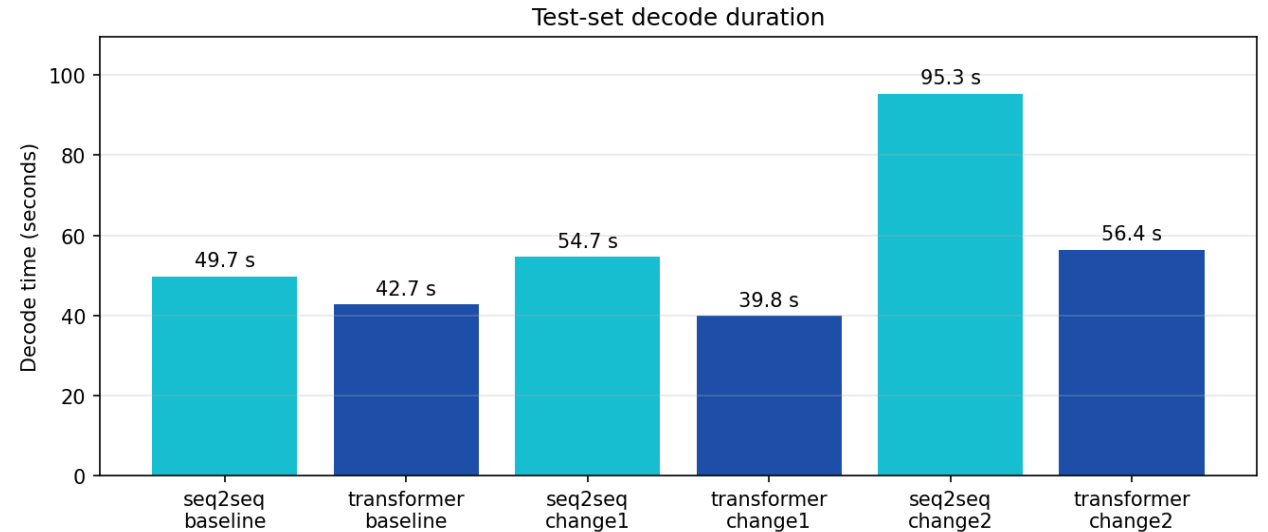
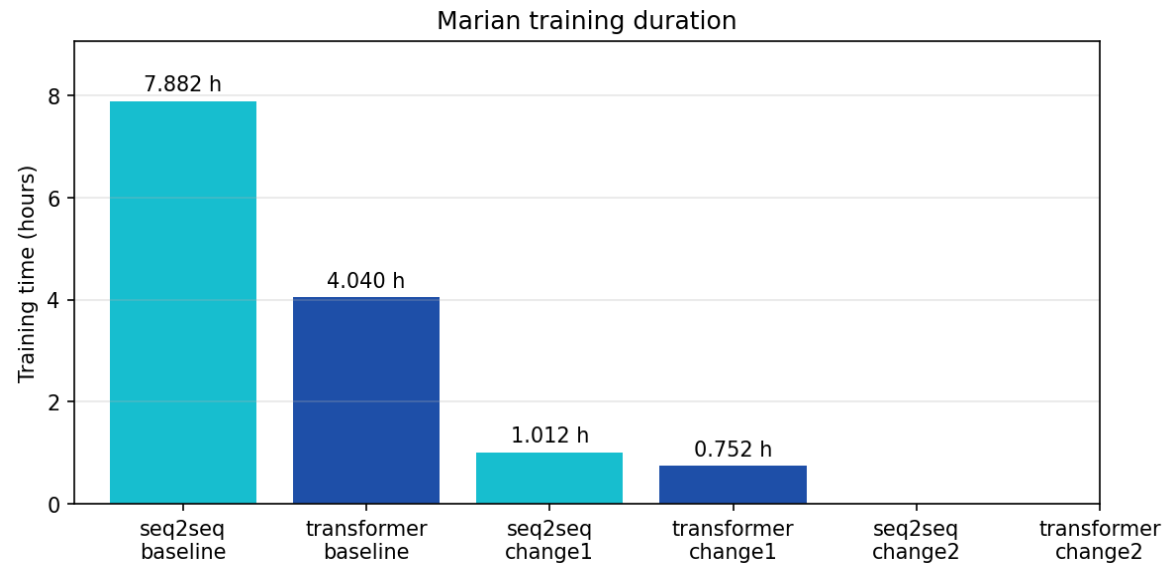
Multi-BLEU on Test dataset across runs

- Seq2Seq
 - baseline:
 - BLEU = 69.49, 85.9/74.0/64.5/57.0 (BP=1.000, ratio=1.000, hyp_len=8044, ref_len=8048)
 - change1, set *--valid-freq* & *--save-freq* to 1000 from 5000; *--early-stopping* 5 from 10:
 - BLEU = 70.31, 86.1/74.5/65.8/58.1 (BP=0.999, ratio=0.999, hyp_len=8043, ref_len=8048)
 - change2, pass *--models model.iter7000.npz model.iter8000.npz* with *--weights 1 1* to Marian decoder:
 - BLEU = 70.44, 86.2/74.6/65.9/58.2 (BP=0.999, ratio=0.999, hyp_len=8042, ref_len=8048)
- Transformer
 - baseline:
 - BLEU = 72.44, 87.6/76.6/67.6/61.0 (BP=0.999, ratio=0.999, hyp_len=8041, ref_len=8048)
 - change1, set *--valid-freq* & *--save-freq* to 1000 from 5000; *--early-stopping* 5 from 10:
 - BLEU = 71.49, 87.2/75.7/66.5/60.0 (BP=0.998, ratio=0.998, hyp_len=8034, ref_len=8048)
 - change 2, pass *--models model.iter4000.npz model.iter5000.npz model.iter6000.npz* with *--weights 1 1 1* to Marian decoder:
 - BLEU = 72.35, 87.4/76.3/67.3/61.4 (BP=0.999, ratio=0.999, hyp_len=8036, ref_len=8048)

Multi-BLEU on Test dataset across runs



Training vs Decode Duration



Details

Data Preprocessing

- Move assignment data files under **data/g2p-par/**
- Syllable-normalization with **syl_normalizer.py** (Version 0.6) on **data/g2p-par/train.my**, **../dev.my**, and **../test.my**
- Create **data/cleaned** for normalized .my files

Seq2Seq baseline

- Modify paths in **seq2seq.mypsh.sh**
- Train
- Decode with the best model (model.iter15000.npz)
- Evaluate BLEU score with **multi-bleu.perl** from Moses

Seq2Seq change1

- Modify **seq2seq.mypsh**
 - *--valid-freq 5000 --save-freq 5000* → *--valid-freq 1000 --save-freq 1000*
 - *--early-stopping 10* → *--early-stopping 5*
- Train
- Decode with the best model (model.iter8000.npz)
- Evaluate BLEU score with **multi-bleu.perl** from Moses

Seq2Seq change2

- Decode with two models around the peak (model.iter7000.npz, model.iter8000.npz)
 - !marian-decoder --models model.iter7000.npz model.iter8000.npz --weights 1 1 ...
 - Averaging checkpoints around the peak is impossible with 3 models because of GPU and RAM limitation!
- Combine the results
- Evaluate BLEU score with **multi-bleu.perl** from Moses

Transformer baseline

- Modify paths in **transformer.mypsh.sh**
- Train
- Decode with the best model (model.iter10000.npz)
- Evaluate BLEU score with **multi-bleu.perl** from Moses

Transformer change1

- Modify **transformer.mypsh.sh**
 - *--valid-freq 5000 --save-freq 5000* → *--valid-freq 1000 --save-freq 1000*
 - *--early-stopping 10* → *--early-stopping 5*
- Train
- Decode with the best model (model.iter5000.npz)
- Evaluate BLEU score with **multi-bleu.perl** from Moses

Transformer change2

- Decode with three models around the peak (model.iter7000.npz, model.iter8000.npz)
 - !marian-decoder --models model.iter4000.npz model.iter5000.npz model.iter6000.npz --weights 1 1 1 ...
- Combine the results
- Evaluate BLEU score with **multi-bleu.perl** from Moses

Thank you!!!